

# Familiarizing with AI

**Session 1** : Introduction to AI and Its Impact

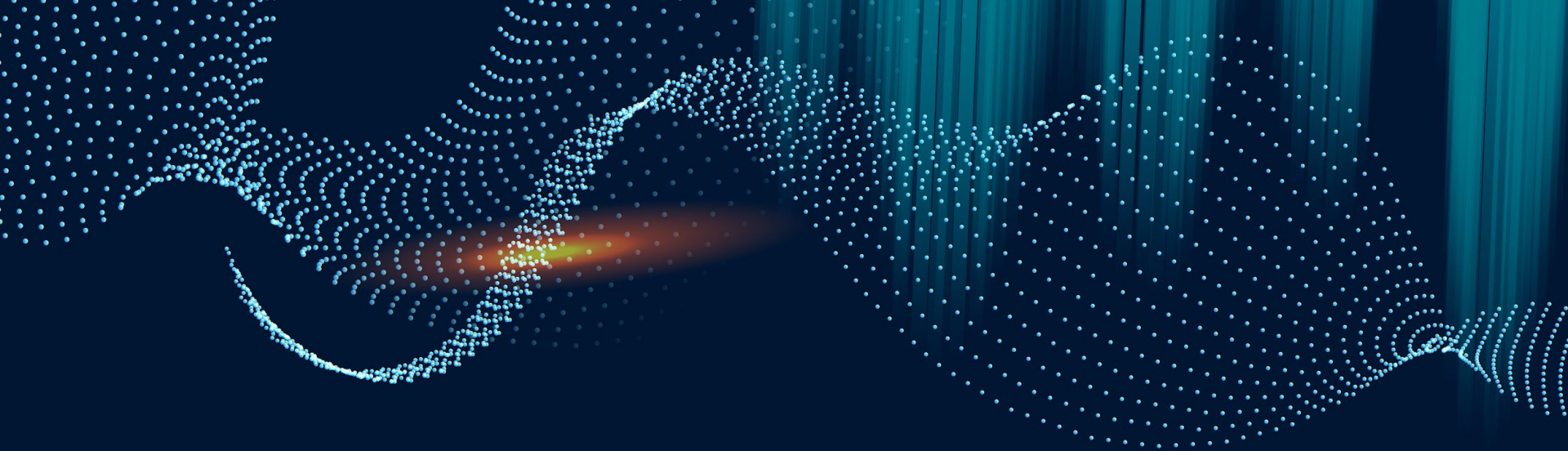


# Introduction to AI and Its Impact

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01

What is AI?

# What is AI?

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Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are designed to think and act like humans:

- understanding natural language
- recognizing patterns
- solving problems
- making decisions



# Core Concepts of AI

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## **Machine Learning (ML)**

A subset of AI that enables machines to learn from data without being explicitly programmed.

## **Deep Learning**

A specialized form of ML using neural networks to model and understand complex patterns, often applied in image and speech recognition.

## **Natural Language Processing (NLP)**

The ability of AI to understand, interpret, and generate human language.

## **Computer Vision**

AI's ability to interpret and understand visual information from the world, like recognizing faces or objects.





# Types of AI

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## **Narrow AI**

AI that is designed to perform a specific task (e.g., chatbots, recommendation engines).

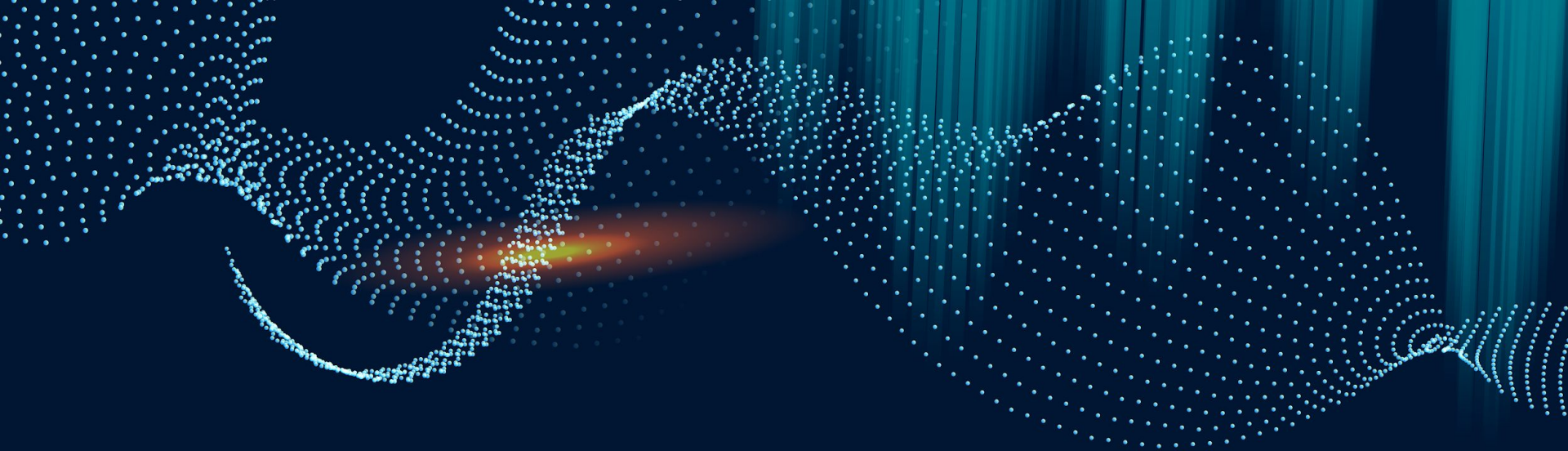
## **General AI**

A theoretical form of AI that can perform any intellectual task that a human can do.

## **Superintelligent AI**

Hypothetical AI that surpasses human intelligence.





02

AI in everyday life

# Applications of AI

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## **Healthcare**

AI aids in diagnostics, personalized treatment plans, and drug discovery.

## **Finance**

AI is used for fraud detection, risk assessment, and trading.

## **Retail**

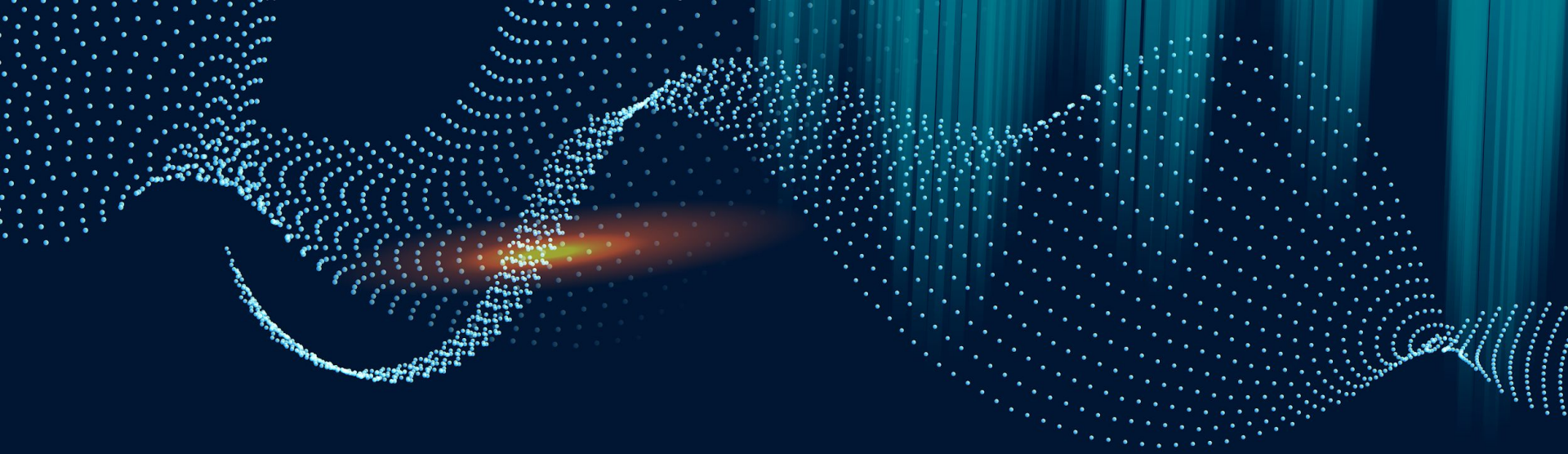
AI powers recommendation systems, inventory management, and personalized marketing.

## **Marketing**

AI is used in predictive analytics and chatbots.







03

Current Trends

# Current Trends

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## **Generative AI**

Tools like GPT-4, DALL-E creating text, art, and more

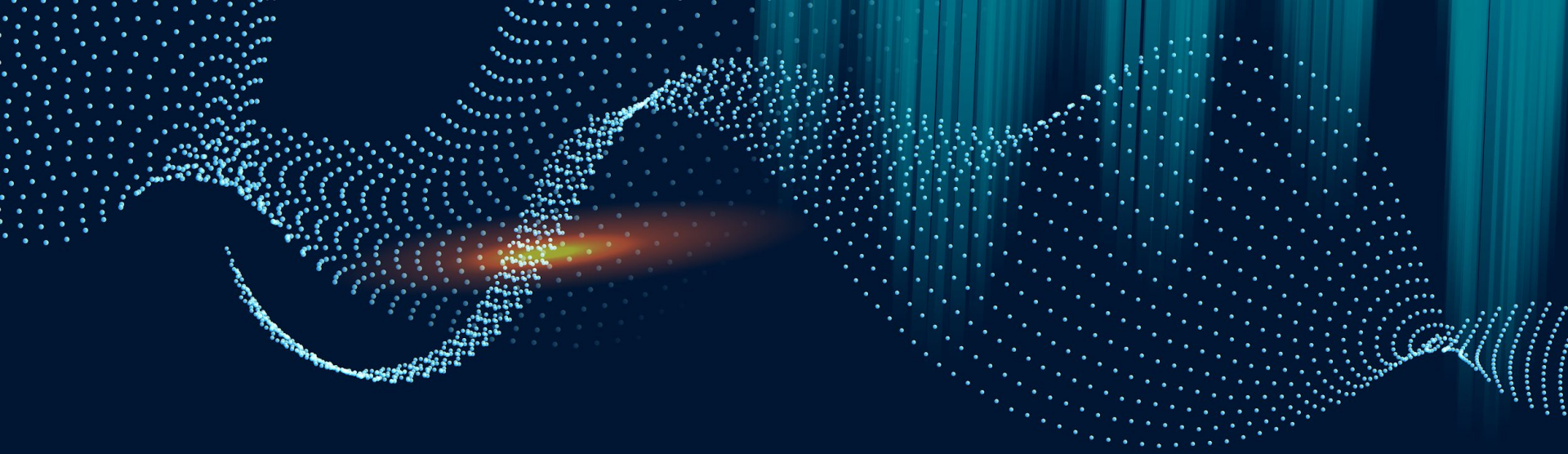
## **Autonomous Systems**

Self-driving cars, drones

## **Environmental Sustainability**

Reducing carbon footprint, climate modeling





04

## Ethical Implications

# Ethical Implications & Challenges

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## **Bias and Fairness**

AI systems can perpetuate and even amplify biases present in the data used to train them which can lead to discrimination in areas like hiring.

## **Privacy and Security**

AI relies heavily on large datasets, often involving sensitive personal information. This raises concerns about how that data is collected, stored, and used, and whether individuals' privacy rights are respected.

## **Job Displacement**

Workers in roles that are easily automated may face unemployment or the need for retraining. While AI may create new jobs, there is a risk that it will eliminate more than it creates, especially in the short term.





# Ethical Implications & Challenges

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## **Accountability & Transparency**

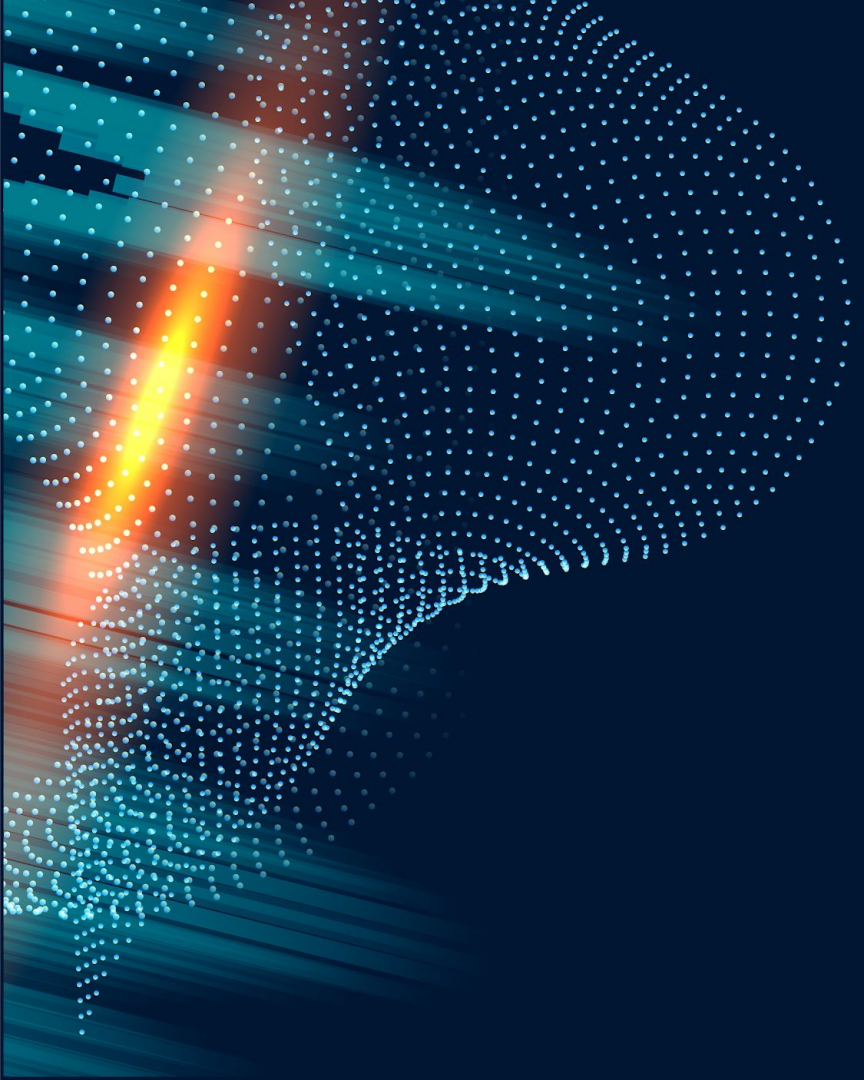
Many AI systems are often referred to as "black boxes" because it is difficult to understand how they arrive at specific decisions. Who is responsible when AI makes a wrong or harmful decision?

## **Ethical Use of Data**

When AI systems use data, it raises questions about whether users have given informed consent and who owns the data. The commercialization of user data by AI companies also raises ethical issues about exploitation and manipulation.





An abstract background featuring a dark blue field. On the left, a bright orange and yellow light streak curves upwards, surrounded by a dense pattern of small white dots that form a semi-circular shape. Several horizontal blue lines with a slight gradient are visible in the upper left quadrant.

**Q&A**